PQRS Homeopathy

(Perfect Quick Reliable Safe)

Submitted in partial fulfillment of the requirements
for the degree of

B.E. Computer Engineering

cc

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2023-2024

CERTIFICATE

This is to certify that the project entitled **"PQRS Homeopathy"** is a bonafide work of **"Joshua Dsouza(A-15), Simson Mastan(A-29) and Steven Mathew(A-44)"** an outhouse project given by **Dr. Rajesh Padghan (Diploma in Homeopathic Medicine and Surgery)** and is submitted to the University of Mumbai in partial fulfillment of the requirement for the award of the degree of B.E. in Computer Engineering.

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TO WHOM IT MAY CONCERN

This is to certify that PQRS Homeopathy & Health Care Services LLP wishes to offer Patient Management System (Homeopathy) to Mr. Steven Mathew, Mr. Simson Mastan and Mr. Joshua D'souza as their final year major project. I would provide them with all necessary details as and when required.

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Project Approval Report for B.E.

This project report entitled ***"PQRS Homeopathy"*** by ***Joshua Dsouza(A-15), Simson Mastan(A-29)and Steven Mathew(A-44)*** is approved for the degree of ***B.E. in Computer Engineering***.

Examiners

1. _____

2. _____

Date:

Place: Mumbai

Declaration

We declare that this written submission represents our ideas in our own words and where others' ideas or words have been included, we have adequately cited and referenced the original sources. We also declare that we have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in our submission. We understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

Signatures of all the students in the group

_____

Joshua Dsouza (A-15)

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Simson Mastan (A-29)

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Steven Mathew (A-44)

Date:

Acknowledgement

We are thankful to a number of individuals who have contributed towards our final year project and without their help; it would not have been possible. Firstly, we offer our sincere thanks to our project guide, Ms. Priya Karunakaran for her constant and timely help and guidance throughout our preparation.We are grateful to all project co-ordinators for their valuable inputs to our project.We are also grateful to the college authorities and the entire faculty for their support in providing us with the facilities required throughout this semester.We are also highly grateful to Dr. Kavita Sonawane, Head of Department (CMPN), Principal, Dr. Sincy George, and Director Bro. Shantilal Kujur for providing the facilities, a conducive environment and encouragement.

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Abstract

Homeopathy, a 200-year-old form of alternative medicine claims to stimulate a healing response and strengthen the body's ability to heal itself. Past COVID-19, Homeopathy has gained its popularity in its disease curing when there were no vaccines produced. Homeopathy also helps in curing several other diseases. In an era marked by rapid digital transformation, the field of healthcare is also experiencing a paradigm shift towards online platforms. This application is designed to provide users with convenient access to homeopathic healthcare from the comfort of their homes. This application leverages cutting-edge technology to connect patients with experienced homeopathic practitioners, ensuring quality healthcare services at their fingertips.

Keywords: *consultation, homeopathy, Wisshom, healthcare, PQRS, case-processing.*

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Chapter 1

Introduction

In the ever-evolving landscape of healthcare, the fusion of technology and traditional medicine is initiating in a new era of accessible and patient-centric care. We are pleased to introduce the PQRS Homeopathy Application, a groundbreaking platform that bridges the gap between centuries-old homeopathic healing and the digital age.

At the heart of our platform lies the invaluable opportunity for users to connect virtually with seasoned homeopathic doctors, fostering a sense of trust and accessibility in their healthcare journey. Beyond mere consultations, our application offers personalized treatment plans tailored to each individual's unique needs and circumstances, ensuring a truly holistic approach to well-being.

At its core, our application is not just about technology; it's about people – individuals seeking comfort, guidance, and solutions for their health concerns. We aim to humanize the experience of accessing homeopathic care, fostering a sense of partnership and empowerment between users and healthcare providers. By bridging the gap between modern technology and traditional healing practices, we aspire to redefine the way individuals engage with and benefit from the world of homeopathy, ultimately paving the way for a healthier, more fulfilling life journey.

1.1 Description

In this comprehensive exploration, we invite you to embark on a journey through the transformative features and numerous benefits of our application. Our aim is not just to showcase technological prowess, but to illuminate the deeply human dimension of healthcare, particularly in the realm of homeopathy.

Imagine being able to consult with experienced homeopathic doctors from the comfort of your own home, feeling truly heard and understood as you discuss your health concerns. These virtual consultations are not just transactions; they're opportunities for genuine connection and collaboration. Together with our practitioners, users co-create treatment plans that resonate with their lifestyles and values, fostering a sense of partnership in their wellness journey.But our commitment to human-centered care doesn't end there. We recognize the importance of support and guidance throughout the healing process. That's why our application includes features like medication reminders, designed to gently nudge users towards better adherence to their treatment plans. Additionally, our wealth of health and wellness resources serves as a beacon of knowledge and inspiration, empowering users to take charge of their health in meaningful ways.

1.2 Problem Formulation

Case processing has been limited to the boundaries of the physical clinic, requiring physical attendance and manual handling of reports by the doctor. This has prevented cross-border interaction of multiple physically unavailable interested patients seeking medical services concurrently, until the availability of the internet and cloud. To facilitate the requirement for patient case processing for the doctor and a concurrently available system for patient interaction and drug delivery, the mission of this project is to provide a distributed system bridging the 2 gaps by providing platform services.

1.3 Motivation

The project helps in a more helpful way to virtually connect to the doctor for having a comfortable consultation with the doctor at the comforts of your home. In this upgrading technology era, getting a chance to show our Engineering skills, the doctor approached to one of us to have us help him for this project.Having an application, brings great help especially for people living around the world as well as people living nearby.

1.4 Proposed Solution

The solution to our problem statement is the PQRS Homeopathy Application. where your health and well-being are at the forefront of our mission to provide holistic healing at your fingertips. This project is being developed in response to the requirements specified by homeopath practitioner Dr. Rajesh Padghan as an outhouse project.

1.5 Scope of the project

The scope for an online homeopathy application is promising, and it encompasses various aspects of healthcare and technology. Here are some key areas where such an application can have a significant scope:

1. Education and Awareness: Offering educational resources on homeopathy, wellness, and holistic living can help users better understand and appreciate this form of healthcare.

2. Global Reach: Expanding the application's reach to different countries and languages can tap into the global interest in holistic healthcare.

3. Regulatory Compliance: Ensuring compliance with healthcare regulations and data privacy laws is essential to build trust among users and avoid legal issues.

4. Research and Development: Conducting research on the effectiveness of homeopathic treatments and continuously improving the application's features can enhance its credibility and user satisfaction.

The scope for an online homeopathy application is not only limited to providing healthcare services but also extends to creating a holistic ecosystem for users to manage their health and well-being. Success in this field would require a combination of technology, medical expertise, user engagement, and regulatory compliance.

Chapter 2

Literature Review

2.1 WissHom Standard

In [1] they have followed the WissHom standard.This documentation standard was published by the German Scientific Society of Homeopathy (WissHom) and was developed in cooperation among three German homeopathy organizations between 2002 and 2008. The guideline includes seven areas: patient information, medical history, homeopathic interview, physical findings, case analysis, prescription, and follow-up monitoring; it includes 16 criteria for documenting case reports.

2.2 Homeopathic Documenting Prescriptions and Clinical Outcomes

In [2] delves into the systematic documentation of prescriptions and clinical outcomes within a homeopathic hospital environment in West Bengal, India.It provides insights into the specific setting of a homeopathic hospital in West Bengal, India, offering a glimpse into the healthcare infrastructure and practices prevalent in that region.The paper likely discusses the methods and procedures employed for documenting prescriptions and clinical outcomes within the hospital. This could include details about the types of records maintained, the format of documentation, and the frequency of updates.It

may elaborate on the clinical outcomes observed as a result of the prescribed homeopathic treatments. This could involve analyzing patient responses to treatment, improvements in health conditions, and any adverse effects encountered.The paper likely presents an analysis of the documented data, highlighting trends, patterns, and correlations between prescriptions and clinical outcomes. This analysis may provide valuable insights into the efficacy and effectiveness of homeopathic treatments in the hospital setting.Based on the findings, the paper may discuss the implications for homeopathic practice and patient care. It may also offer recommendations for improving documentation practices or enhancing patient outcomes based on the observed data.

2.3 Role of Big Data in the Healthcare Sector

In [3] explores the multifaceted role of big data in the healthcare sector, focusing on its management, analysis, and potential future directions.The paper likely begins by introducing the concept of big data and its relevance in the healthcare industry. It may discuss the vast amounts of data generated from various sources such as hospital records, medical examinations, and biomedical research.It may delve into the challenges associated with managing and analyzing big data in healthcare, highlighting issues such as data volume, variety, velocity, and veracity. Additionally, the paper may explore the opportunities presented by big data analytics in improving healthcare services and patient outcomes.The paper likely discusses strategies and technologies for effectively managing big data in healthcare settings. This may include topics such as data storage, retrieval, security, and privacy measures to ensure compliance with regulatory standards such as HIPAA.It may delve into various analytical techniques and methodologies used to extract meaningful insights from healthcare big data. This could involve discussing data mining, machine learning, natural language processing, and predictive analytics as applied to healthcare

datasets.The paper may present case studies or examples illustrating real-world applications of big data analytics in healthcare. This could include examples of how big data has been used to improve clinical decision-making, disease surveillance, personalized medicine, and population health management. Finally, the paper may speculate on the future prospects of big data in healthcare, discussing emerging trends, technological advancements, and potential challenges on the horizon. It may also suggest areas for further research and development in the field of healthcare analytics.

2.4 The Applicability and Effectiveness of the modified Naranjo criteria

In [4], the study evaluates the applicability and effectiveness of the modified Naranjo criteria in attributing causality of clinical outcomes to homeopathic interventions, as presented in case reports. Utilizing a retrospective analysis approach, a cohort of case reports was reviewed, and the modified Naranjo criteria were applied to assess the likelihood of causality between homeopathic treatment and observed clinical outcomes. The study aimed to provide insights into the validity and reliability of using such criteria within the context of homeopathic practice.Homeopathy, a complementary and alternative medicine system, relies on individualized treatment approaches based on the principle of "like cures like." While case reports serve as valuable sources of clinical evidence in homeopathy, assessing causality between treatment and outcomes can be challenging. The Naranjo criteria, originally developed for pharmacovigilance, have been adapted for use in assessing causal attribution in homeopathic case reports. This study seeks to evaluate the effectiveness of these modified criteria in this context.The findings suggest the potential utility of the modified Naranjo criteria in assessing causal attribution in homeopathic case reports. However, further refinement and validation of these criteria are warranted to enhance their

reliability and applicability in clinical practice. Future research should focus on standardizing assessment methodologies and incorporating complementary approaches to strengthen the evidence base for homeopathy.

2.5 Design and Implementation of a Family Doctor Application

in [5], the study presents the design and implementation of a family doctor application developed specifically for the Android platform. Recognizing the growing need for accessible healthcare services, especially in the context of family health management, the app aims to provide convenient access to medical resources, health information, and personalized healthcare services. Leveraging the capabilities of the Android platform, the app offers features such as appointment scheduling, health record management, medication reminders, and telemedicine consultations. The study highlights the development process, key features, and potential benefits of the family doctor app in improving healthcare accessibility and patient outcomes.

2.6 AI-based Medical Assistant Chatbots

In [7], the review article provides an overview of AI-based medical assistant chatbots, focusing on their applications, capabilities, and potential impact on healthcare delivery. As the field of artificial intelligence continues to advance, chatbots have emerged as promising tools for enhancing patient engagement, providing personalized health information, and supporting healthcare professionals in clinical decision-making. The review examines various AI techniques employed in medical chatbots, discusses their strengths and limitations, and highlights key considerations for their implementation and integration into healthcare systems.

Chapter 3

System Analysis

3.1 Functional Requirements:

1. User Registration and Authentication: Users can create accounts securely through registration and authenticate themselves using credentials like email and password, ensuring data privacy and access control.

2. Disease Selection: Patients select their specific illnesses or conditions from a predefined list, facilitating personalized healthcare management and treatment.

3. Questionnaire Generation: Upon disease selection, the system dynamically generates questionnaires tailored to the selected illness, gathering relevant information for diagnosis and treatment planning.

4. User Questionnaire Interaction: Patients interact with the generated questionnaires, providing responses that aid in symptom assessment, medical history collection, and treatment recommendation.

5. Data Submission: Patient responses from the questionnaires are submitted securely to the system, where they are processed and stored for further analysis and review.

6. Doctor Review: Medical professionals review the submitted patient data, including symptoms, medical history, and questionnaire responses, to

make informed diagnostic and treatment decisions.

7. Medicine Prescription: Based on the doctor's review, appropriate medications or treatments are prescribed to patients, considering factors such as disease severity, patient condition, and drug interactions.

8. Patient Profile: Each patient has a profile within the system containing their personal information, medical history, treatment records, and prescribed medications, facilitating continuity of care and treatment tracking.

3.2 Non-functional Requirements:

1. Security: The system employs robust encryption protocols, access controls, and regular security audits to safeguard sensitive patient data and ensure compliance with data protection regulations, mitigating risks of unauthorized access or data breaches.

2. Performance: Through efficient database management, optimized code, and scalable infrastructure, the system ensures fast response times, minimal downtime, and seamless user experience, even during periods of high user activity, enhancing overall system performance and reliability.

3. Usability: User-centric design principles, intuitive interfaces, and comprehensive user training programs are implemented to enhance the system's ease of use, enabling both patients and healthcare professionals to navigate the platform effortlessly and maximize productivity.

4. Scalability: The system architecture is designed to seamlessly accommodate growing user bases, increasing data volumes, and expanding feature sets, ensuring scalability to meet evolving healthcare demands without compromising performance or user experience.

5. Compliance: Adherence to industry standards, regulatory requirements, and best practices guides the system's development and operation, ensuring compliance with healthcare regulations such as HIPAA, GDPR, and local data protection laws, thereby mitigating legal risks and fostering trust among users.

3.3 Specific Requirements

- Questionnaire Customization
- Medicine Database
- Feedback and Review
- Cross-Platform Compatibility

3.4 Use Case Diagram and Specifications

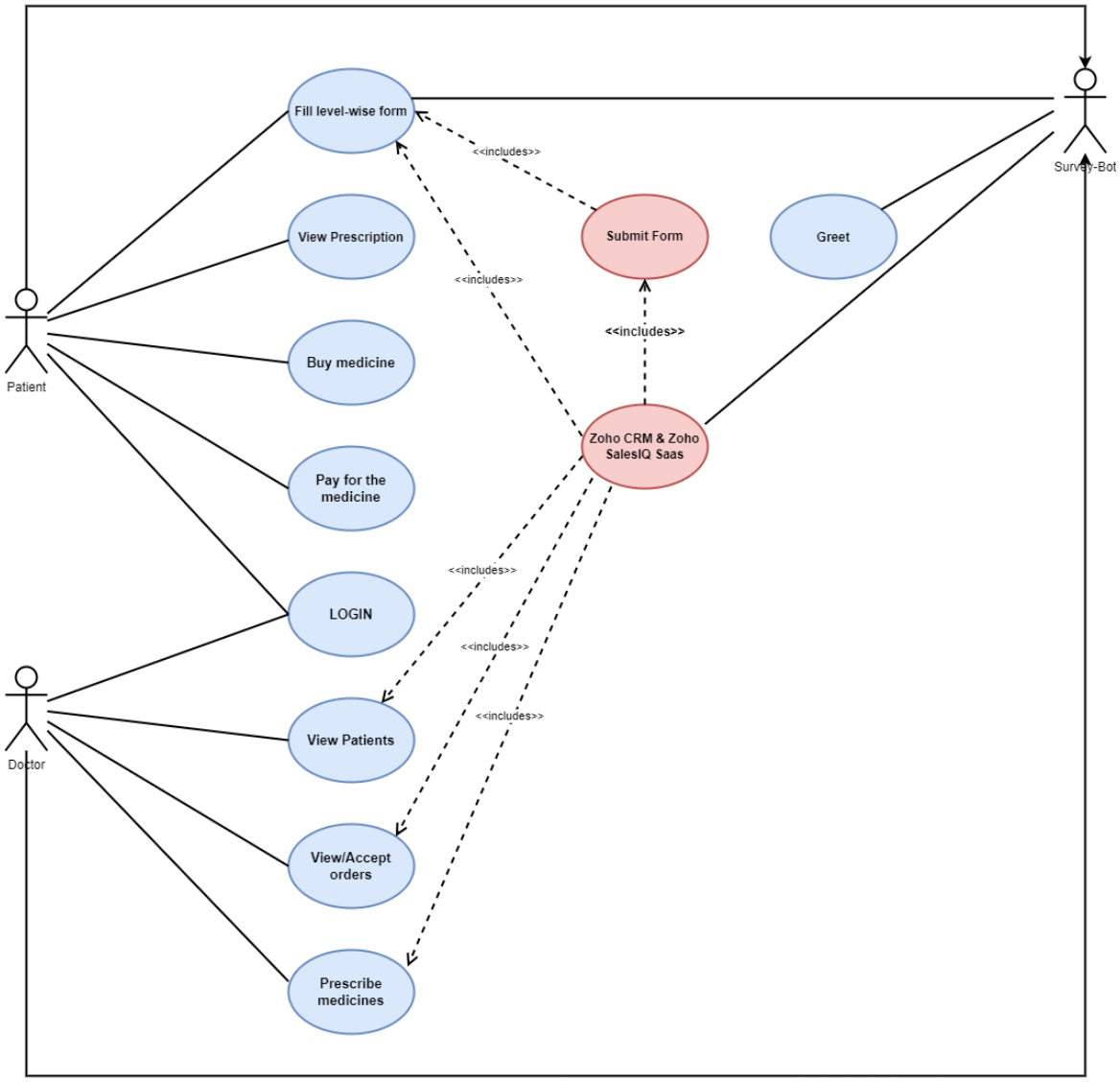


Figure 3.1: Use-case Diagram

Use Case 1: Fill-level wise forms
Actor: Patient
Description: In this activity the patient has to fill level-wise forms based on the disease he/she has selected.
Pre-Condition: The frontend has loaded
Post-Condition: The user has to fill the forms appeared

Use Case 2: Buy medicine
Actor: Patient
Description: Based on the prescription provided after the verification from the doctor, the patient has to purchase the medicine prescribed.
Pre-Condition: Doctor receives the response
Post-Condition: **The user** has to purchase the medicine prescribed.

Use Case 3: Pay for the medicine
Actor: Patient
Description: In this activity the patient has to pay for the medicine purchased.
Pre-Condition: The cost of the medicine will be displayed below its name
Post-Condition: User can track once ordered.

Use Case 4: Login
Actor: Doctor
Description: In this activity the doctor will login to the application
Pre-Condition: The doctor will receive notification of a response arrived
Post-Condition: Based on the response the doctor will prescribe medicines.

Use Case 5: View Patients
Actor: Doctor
Description: In this activity the doctor has full access to view the number

of patients, their responses and their orders.
Pre-Condition: Has to be in logged into the application.
Post-Condition: Patients are verified.

Use Case 6: View/Accept orders
Actor: Doctor
Description: In this activity the doctor accepts the order placed by the patient based on the prescription he sends.
Pre-Condition: The patient orders for the medicine prescribed.
Post-Condition: The order is placed and the medicine reaches its location based on the place to be delivered.

Use Case 7: Prescribe medicines
Actor: Doctor
Description: In this activity the doctor prescribes medicines based on the responses sent by the patient.
Pre-Condition: The patient has to submit the form for the disease selected.
Post-Condition: Prescription is sent to the patient and is viewd by them.

Use Case 8: Greet
Actor: Survey-Bot
Description: In this activity the bot interacts with the patients discussing deeply about his/her disease.
Pre-Condition: The patient has to select a disease.
Post-Condition: Bot talks and the submission is viewd by the doctor.

Chapter 4

Analysis Modelling

4.1 Activity Diagram

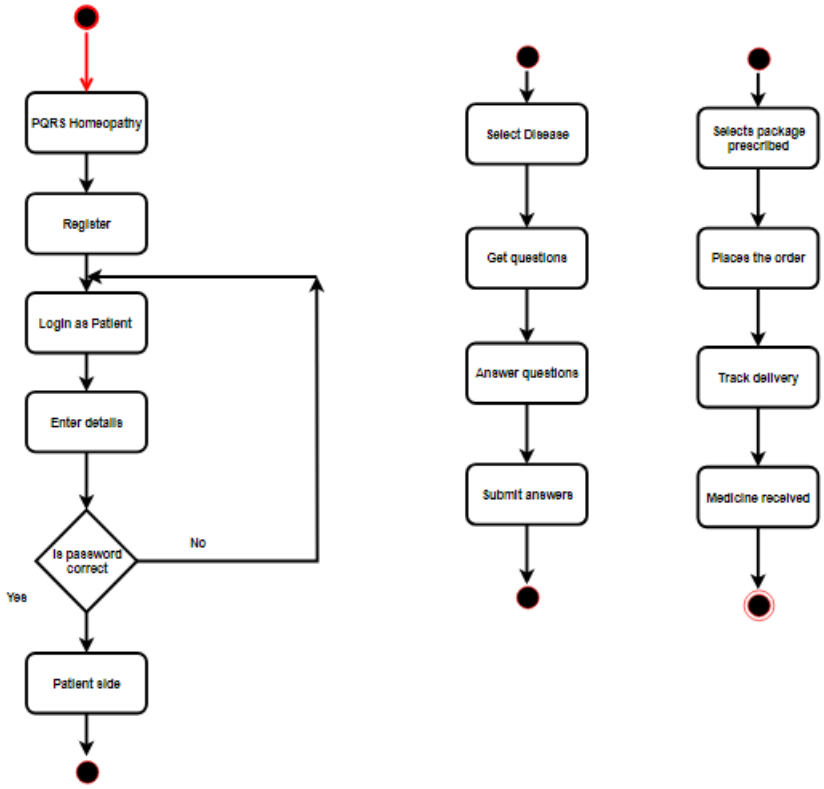


Figure 4.1: Activity diagram for Patient

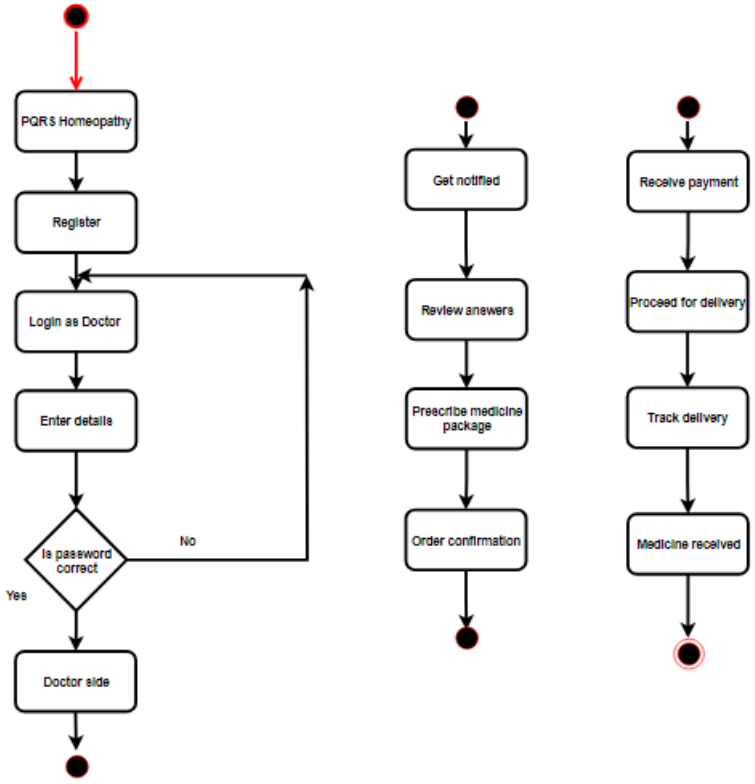


Figure 4.2: Activity diagram for Doctor

Activity diagram (Figure 4.1 and 4.2) describes the activity of the application. It begins from registration by both the patient as well as the doctor. The patient here enters his credentials and when he/she selects the disease, based on the selected disease, a set of questions appear to the patient as various forms.The patient has to answer the questions given in the form and has to submit the form. The form once submitted, the doctor gets notified and he goes through the responses based on which he prescribes the patient what medicine package he should select for his/her disease. Prescription once viewed by the patient, he/she buys the prescribed package, select the method of payment and then places the order. Once the doctor receives the order confirmation, he proceeds the delivery process of the medicine. The patient as well as the doctor can track the delivery and the medicine is delivered to the patient.

4.2 Data flow diagram

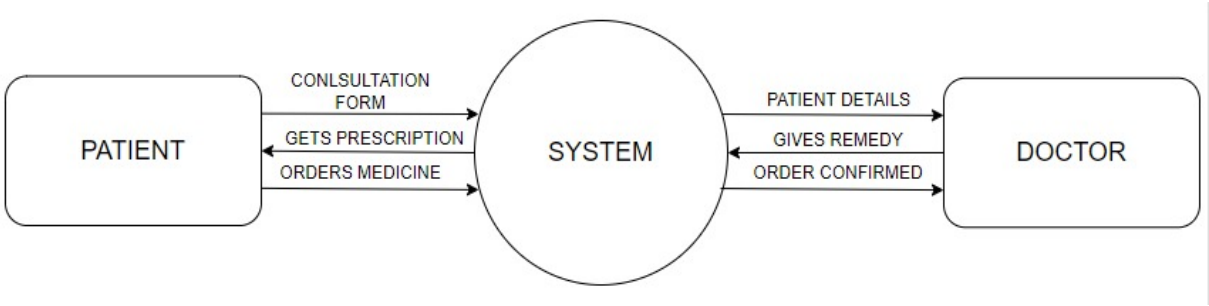


Figure 4.3: Data flow diagram level-0

The above diagram (Figure no. 4.3) consists of the Level 0 DFD which is a way of representing a flow of data through a process or a system. The above diagram provides an overview of the entire system, outlining the high-level processes and interactions between the system and external entities involved in the logistics and fabrication processes. This serves as the foundation for more detailed DFDs to describe the system's internal processes and functionalities in subsequent levels.

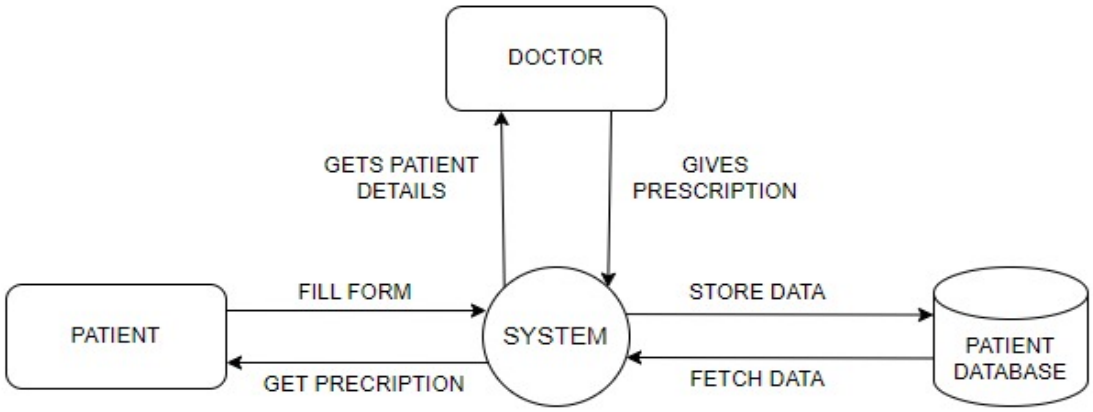


Figure 4.4: Data flow diagram level-1

The above diagram (Figure no. 4.4) shows the Level 1 Data Flow Diagram (DFD) for PQRS Homeopathy. It shows the system's functionalities and data interactions within the logistics and fabrication processes. The diagram features distinct processes. The Patient Order Management process involves receiving and processing customer orders through the website interface. It verifies and validates the order details before storing the data in the Orders Database. The Inventory Management process monitors inventory levels, receives restocking alerts, and updates the Inventory Database accordingly. The E-store Interaction process manages the communication and tracking of incoming medicine orders from patient, recording details in the Orders Datebase.

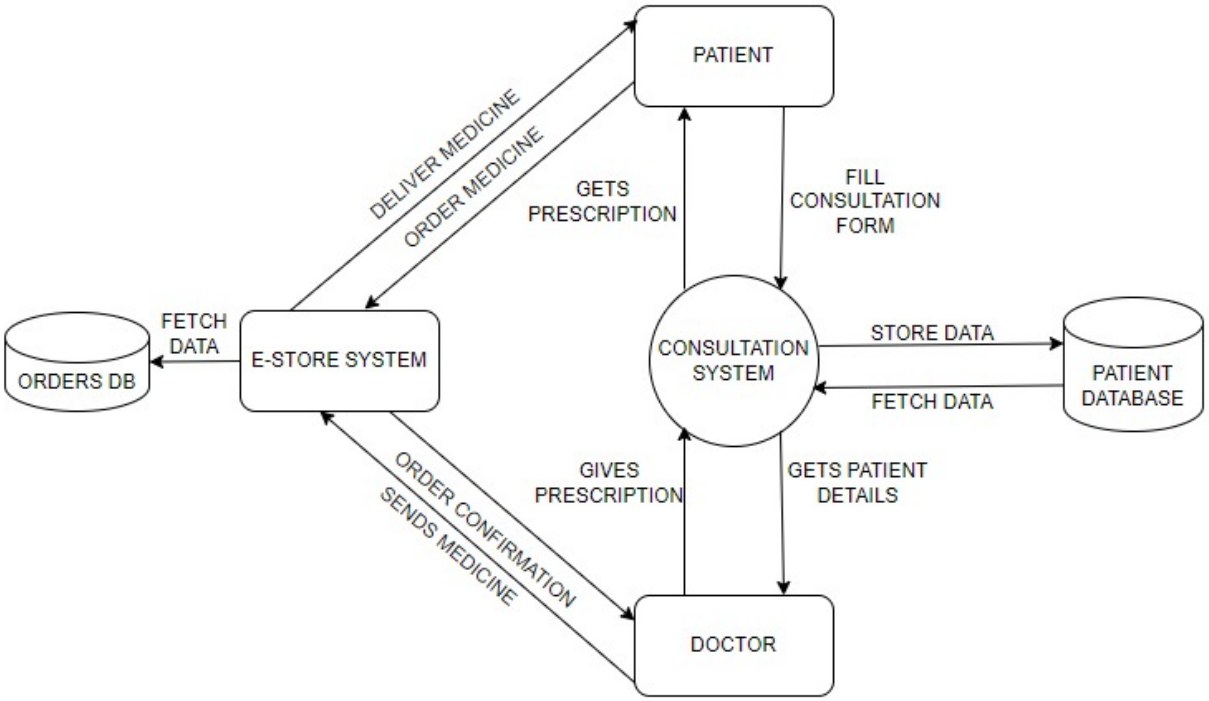


Figure 4.5: Data flow diagram level-2

In the Level 2 Data Flow Diagram (DFD) for PQRS Homeopathy, the detailed depiction showcases a more granular view of the system's functionalities, specifically focusing on the website's management involving patient interactions, system processes, and medicine supply. This diagram elaborates on the Level 1 processes, presenting specific sub-processes and their interrelations. The Patient Order Management process delves into multiple detailed steps involving order validation, verification, and updating the Orders Database with the order details. The Consultation Management process handles patient details, patient illness records, and updates to the Patient Database, illustrating the flow within the system. The E-store system process oversees medicine ordering process with patient, recording patient details, and updating the Order Database.

4.3 Entity Relationship Diagram

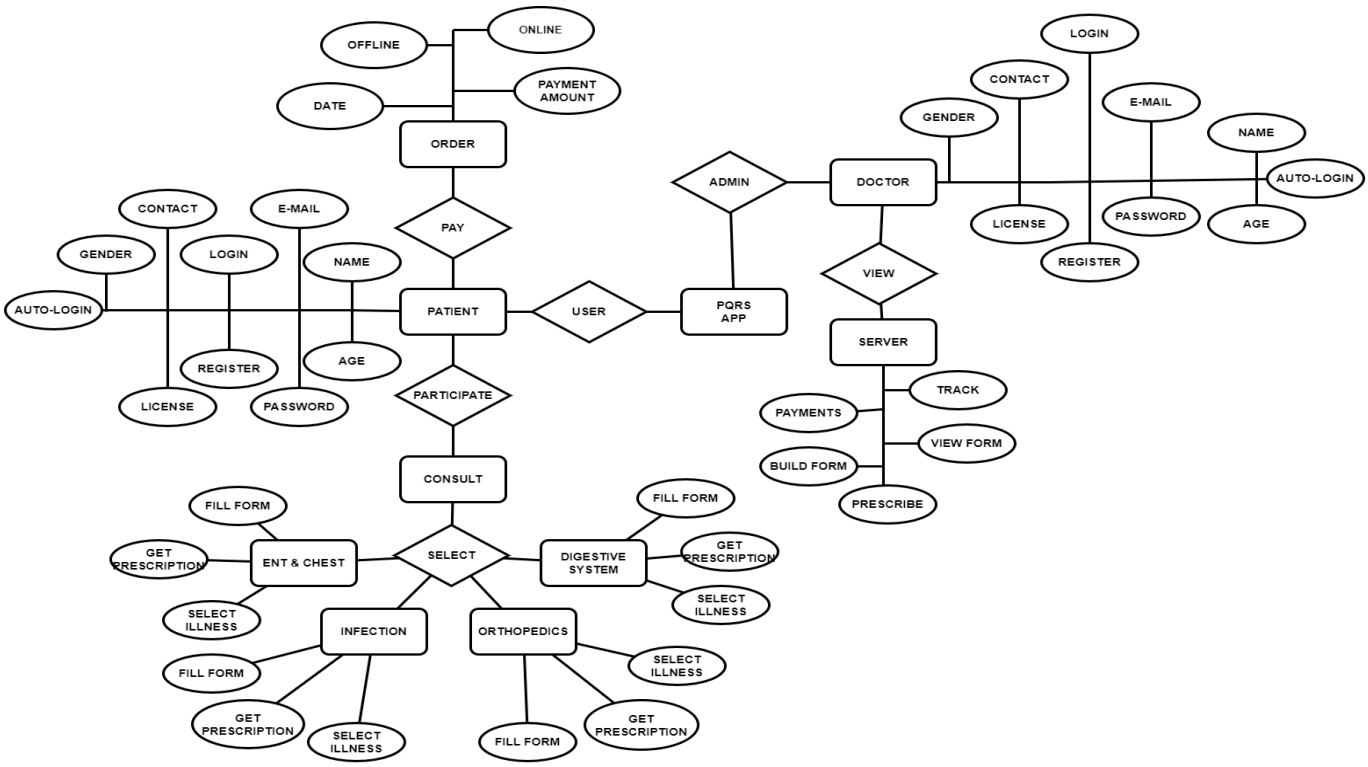


Figure 4.6: Entitity Relationship Diagram

In this diagram, the entire flow of the PQRS Homeopathy application is explained. From the beginning of the application, from the patients' entry point to the doctors' side can be see from the above diagram. The figure shows how a patient registers, logins, interacting with a small navigation chat, making his/her navigation to his/her problem in a faster way.A patient selects the disease and then chooses his/her specific illness based on which he/she has to fill a form which contains questions based on the illness and once the form is submitted, the doctor receives and reads the responses based on which he will prescribe medicines to the patient and selection of the package is done by the patient and once payment mode confirmed, the medicine is delivered to the patient.

4.4 Timeline Chart

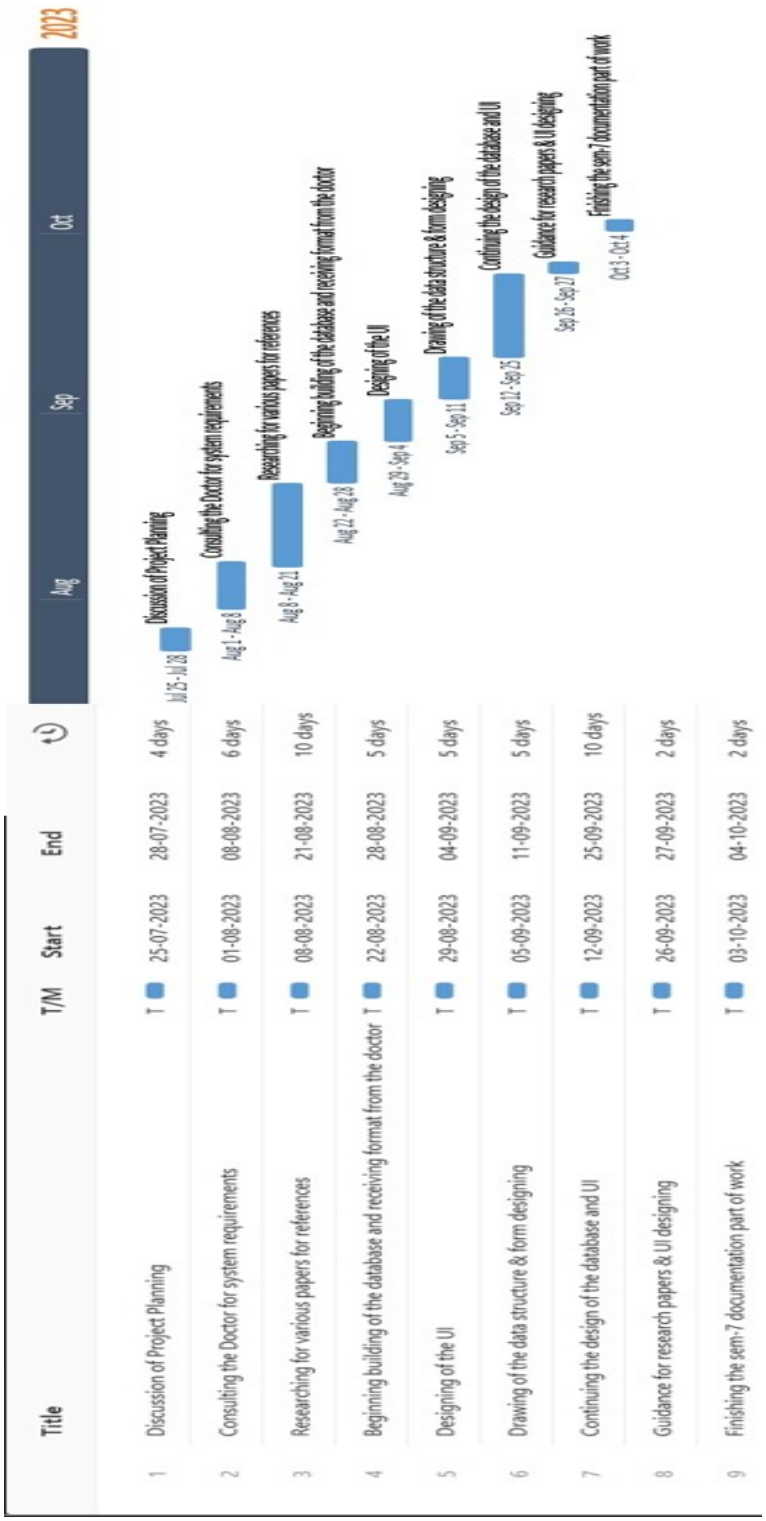


Figure 4.7: Timeline Chart (Sem-07)

4.5 Timeline Chart

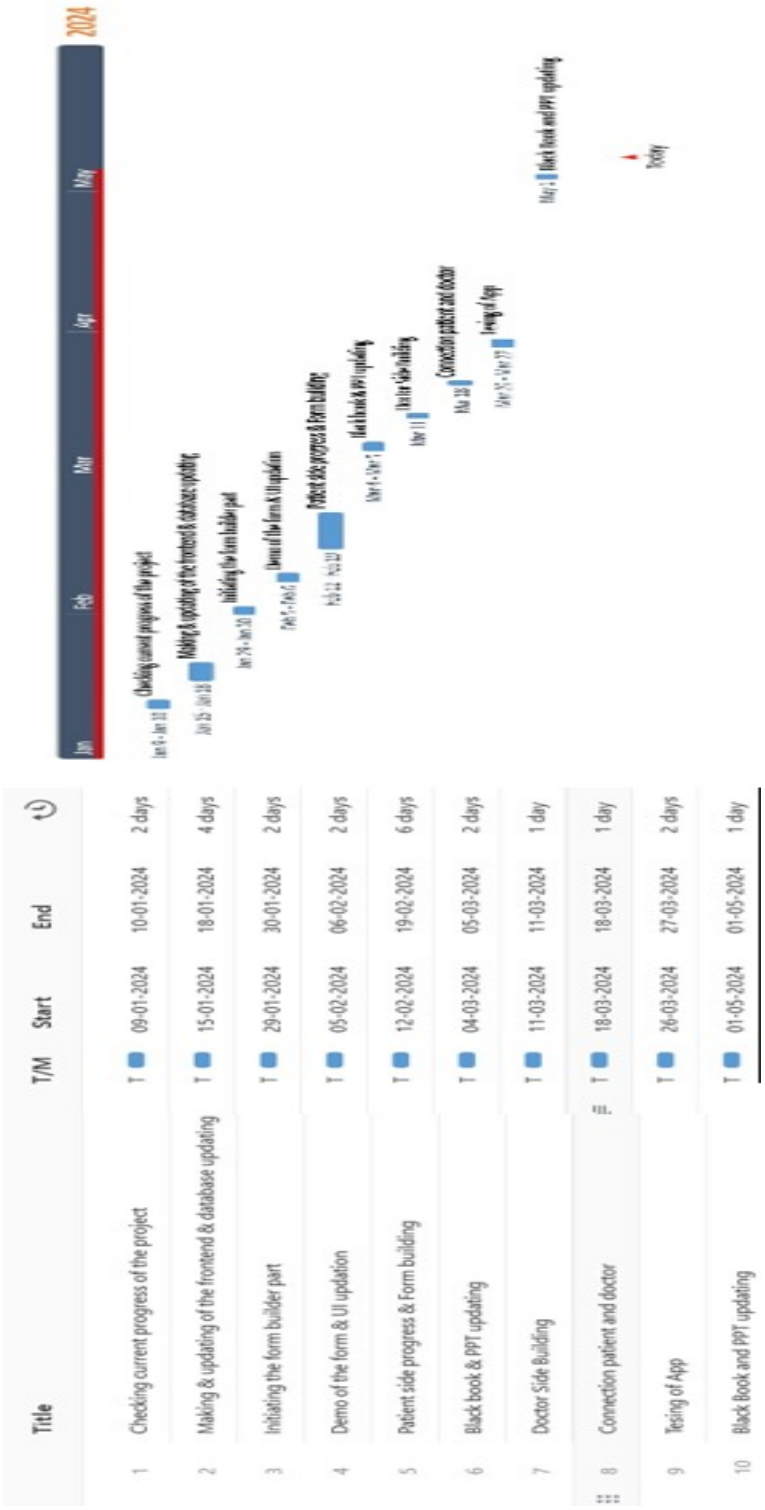


Figure 4.8: Timeline Chart (Sem-08)

Chapter 5

Design

5.1 Architectural Design

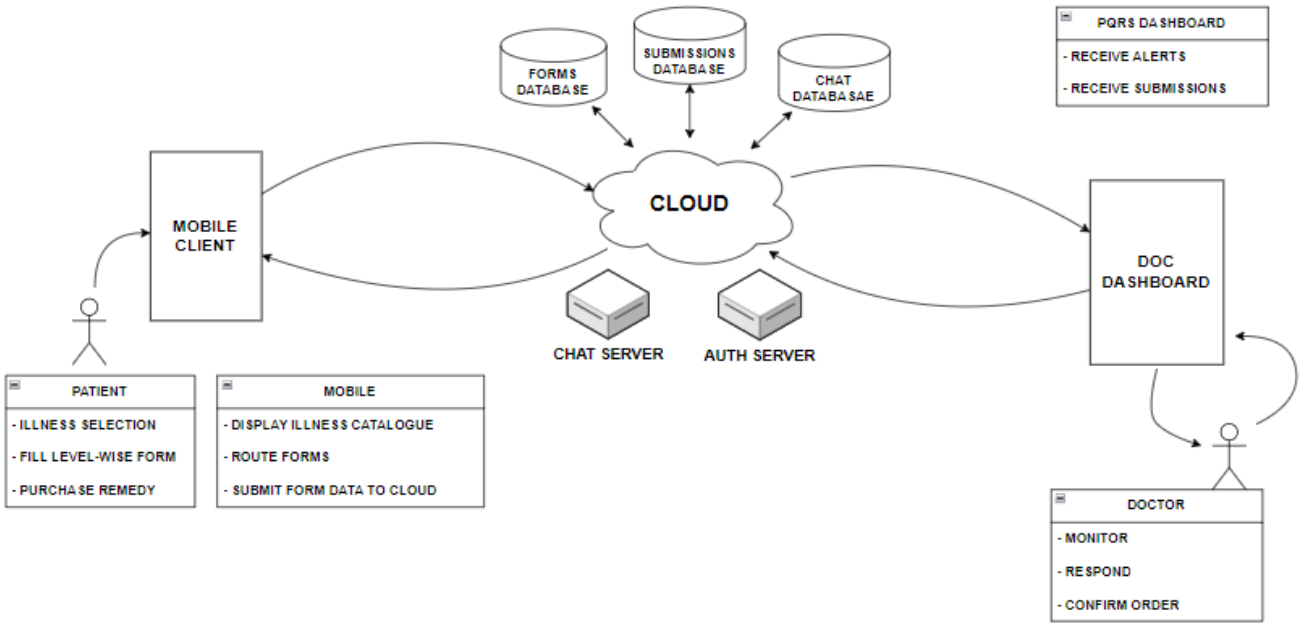


Figure 5.1: Architectural Design

The architecture diagram you described follows a logical sequence of actions within the application, outlining the workflow and interaction between patients and doctor.

- Patient Registration and Symptom Submission:

  When a patient opens the application, they are required to register or log in. After successful registration or login, the patient starts a level-wise form. In this form, the patient fills in their personal details, such as name, age, contact information, and medical history. Additionally, the patient describes the symptoms and illnesses they are currently experiencing, providing the doctor with comprehensive information.

- Form Submission and Doctor's Reception:

  Once the patient submits the form, the data is sent to the server or a database where it can be accessed by authorized medical professionals. The doctor, who has access to the application, is notified of the new patient submission. The patient's data is presented to the doctor in a structured manner, allowing for efficient review and analysis.

- Doctor's Assessment and Prescription:

  The doctor reviews the patient's submitted information, including their personal details and reported symptoms and illnesses. Based on their assessment, the doctor recommends the appropriate medicine and treatment plan for the patient's condition. The prescription is generated within the application and shared with the patient electronically.

- Patient's Access to Prescription:

  The patient receives the prescription through the application, providing them with details about the prescribed medicine, dosage, and treatment instructions. Having access to the prescription within the app ensures that the patient has a digital record of their treatment plan.

- Medicine Ordering:

  Patients can order the recommended medicine directly through the application. The app may provide features for browsing available medications, viewing their details, and placing an order. Ordering through the app streamlines the process and provides convenience to the patient.

- Doctor-Patient Interaction:

  If the doctor has any additional instructions or needs to communicate with the patient, the application may facilitate secure messaging or telehealth features for real-time interaction.After the patient places an order, the doctor or a pharmacy prepares the medication.The medicine is then delivered to the patient through an offline delivery process, ensuring they receive the prescribed treatment.

Chapter 6

Implementation

6.1 Algorithms/Methods Used

The algorithms and methods used in a project can vary greatly depending on its specific requirements, objectives, and technologies involved. However, in our PQRS Homeopathy App, some common algorithms and methods that have been employed are:

1. Data Retrieval and Storage: Firebase Realtime Database is used for storing user data, consultation forms, orders, etc., and retrieving them in real-time.Firebase Authentication is For user authentication and ensuring secure access to the app's features.

2. Data Processing and Analysis: Parsing JSON data retrieved from Firebase to extract relevant information like user details, orders, etc.

3. User Interaction and Interface: RecyclerView and Adapters are utilized for displaying lists of users, their orders, and consultation forms in the app's UI efficiently. Button Click Handlers are used for andling button clicks to perform actions like accepting orders, sending prescriptions, etc.

4. Prescription Generation: Implementing rules and heuristics to generate prescriptions based on consultation forms and symptoms provided by users.Utilizing decision trees or expert systems to guide the prescription generation process based on homeopathic principles and medical knowledge.

5. Communication and Notifications: Sending push notifications to users to notify them about prescription updates, order acceptance, etc.

6. Security: Implementing encryption and hashing techniques to secure sensitive user data stored in the database and transmitted over the network.

7. User Experience Optimization: Employing techniques to optimize app performance, such as caching frequently accessed data, minimizing network calls, etc. Conducting usability testing to refine the app's user interface and ensure a seamless user experience.

6.2 Working of the project

1. PatientRegistration.Java

```
1  package com.example.finalehomeopathy;
2  import android.content.Intent;
3  import android.os.Bundle;
4  import android.text.TextUtils;
5  import android.util.Log;
6  import android.view.View;
7  import android.widget.EditText;
8  import android.widget.RadioButton;
9  import android.widget.Toast;
10
11 import androidx.annotation.NonNull;
12 import androidx.appcompat.app.AppCompatActivity;
13
14 import com.google.firebase.auth.FirebaseAuth;
15 import com.google.firebase.auth.FirebaseUser;
16 import com.google.firebase.database.DatabaseReference;
17 import com.google.firebase.database.FirebaseDatabase;
18
19 public class PatientRegistrationActivity extends AppCompatActivity {
20
21     private EditText editTextName, editTextAge, editTextEmail, editTextPasswor
22     private RadioButton radioButtonMale, radioButtonFemale;
23     private FirebaseAuth mAuth;
```

```
24      private DatabaseReference mDatabase;
25
26      @Override
27      protected void onCreate(Bundle savedInstanceState) {
28          super.onCreate(savedInstanceState);
29          setContentView(R.layout.activity_patient_register);
30
31          mAuth = FirebaseAuth.getInstance();
32          mDatabase = FirebaseDatabase.getInstance().getReference();
33
34          editTextName = findViewById(R.id.editText_name);
35          editTextAge = findViewById(R.id.editText_age);
36          editTextEmail = findViewById(R.id.editText_email);
37          editTextPassword = findViewById(R.id.editText_password);
38          editTextConfirmPassword = findViewById(R.id.editText_confirm_password)
39          editTextContactNo = findViewById(R.id.editText_contact_no);
40          radioButtonMale = findViewById(R.id.radioButton_male);
41          radioButtonFemale = findViewById(R.id.radioButton_female);
42
43          findViewById(R.id.button_register_submit).setOnClickListener(new View.
44              @Override
45              public void onClick(View v) {
46                  registerDoctor();
47              }
48          });
49      }
50
51      private void registerDoctor() {
52          String name = editTextName.getText().toString().trim();
53          String age = editTextAge.getText().toString().trim();
54          String email = editTextEmail.getText().toString().trim();
55          String password = editTextPassword.getText().toString().trim();
56          String confirmPassword = editTextConfirmPassword.getText().toString()
57          String contactNo = editTextContactNo.getText().toString().trim();
58          String gender = radioButtonMale.isChecked() ? "Male" : "Female";
59
60          if (TextUtils.isEmpty(name) || TextUtils.isEmpty(age) || TextUtils.isE
61              Toast.makeText(this, "All␣fields␣are␣required", Toast.LENGTH_SHORT
62              return;
63          }
64
```

```
        if (!password.equals(confirmPassword)) {
            Toast.makeText(this, "Passwords do not match", Toast.LENGTH_SHORT)
            return;
        }

        mAuth.createUserWithEmailAndPassword(email, password)
                .addOnCompleteListener(this, task -> {
                    if (task.isSuccessful()) {
                        FirebaseUser user = mAuth.getCurrentUser();
                        String userId = user.getDisplayName();
                        DatabaseReference currentUserDb = mDatabase.child("use
                        currentUserDb.child("name").setValue(name);
                        currentUserDb.child("age").setValue(age);
                        currentUserDb.child("email").setValue(email);
                        currentUserDb.child("gender").setValue(gender);
                        currentUserDb.child("contactNo").setValue(contactNo);

                        Toast.makeText(PatientRegistrationActivity.this, "Regi
                        startActivity(new Intent(PatientRegistrationActivity.t
                        finish();
                    } else {
                        Toast.makeText(PatientRegistrationActivity.this, "Regi
                    }
                })
                .addOnFailureListener(e -> {
                    // Log the exception or show a message to the user
                    Log.e("Registration Error", e.getMessage());
                    Toast.makeText(PatientRegistrationActivity.this, "Registra
                });
    }
}
```

2. PatientRegistration.XML

```
    <?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout xmlns:android="http://schem
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:background="#E1F4F5"
    tools:context=".PatientRegistrationActivity">
```

```
9
10      <ImageView
11          android:id="@+id/image"
12          android:layout_width="154dp"
13          android:layout_height="124dp"
14          android:layout_centerHorizontal="true"
15          android:adjustViewBounds="true"
16          android:maxWidth="180dp"
17          android:src="@drawable/logo"
18          app:layout_constraintEnd_toEndOf="parent"
19          app:layout_constraintStart_toStartOf="parent"
20          app:layout_constraintTop_toTopOf="parent"
21          tools:ignore="MissingConstraints" />
22
23      <TextView
24          android:id="@+id/text1"
25          android:layout_width="match_parent"
26          android:layout_height="wrap_content"
27          android:layout_marginTop="24dp"
28          android:text="Welcome␣to␣PQRS␣Homeopathy"
29          android:textAlignment="center"
30          android:textAppearance="@style/TextAppearance.AppCompat.Headline"
31          android:textSize="20dp"
32          app:layout_constraintEnd_toEndOf="parent"
33          app:layout_constraintStart_toStartOf="parent"
34          app:layout_constraintTop_toBottomOf="@+id/image" />
35
36      <androidx.cardview.widget.CardView
37          android:layout_width="0dp"
38          android:layout_height="0dp"
39          android:layout_marginStart="16dp"
40          android:layout_marginEnd="16dp"
41          android:layout_marginTop="16dp"
42          app:layout_constraintBottom_toBottomOf="parent"
43          app:layout_constraintEnd_toEndOf="parent"
44          app:layout_constraintStart_toStartOf="parent"
45          app:layout_constraintTop_toBottomOf="@+id/text1">
46
47          <ScrollView
48              android:layout_width="match_parent"
49              android:layout_height="match_parent"
```

```
50              android:fillViewport="true">
51
52              <LinearLayout
53                  android:layout_width="match_parent"
54                  android:layout_height="wrap_content"
55                  android:orientation="vertical"
56                  android:padding="16dp">
57
58                  <com.google.android.material.textfield.TextInputLayout
59                      android:id="@+id/nameInputLayout"
60                      android:layout_width="match_parent"
61                      android:layout_height="wrap_content"
62                      android:hint="Name">
63
64                      <com.google.android.material.textfield.TextInputEditText
65                          android:id="@+id/editText_name"
66                          android:layout_width="match_parent"
67                          android:layout_height="wrap_content"
68                          android:inputType="textPersonName" />
69
70                  </com.google.android.material.textfield.TextInputLayout>
71
72                  <com.google.android.material.textfield.TextInputLayout
73                      android:id="@+id/ageInputLayout"
74                      android:layout_width="match_parent"
75                      android:layout_height="wrap_content"
76                      android:hint="Age">
77
78                      <com.google.android.material.textfield.TextInputEditText
79                          android:id="@+id/editText_age"
80                          android:layout_width="match_parent"
81                          android:layout_height="wrap_content"
82                          android:inputType="number" />
83
84                  </com.google.android.material.textfield.TextInputLayout>
85
86                  <com.google.android.material.textfield.TextInputLayout
87                      android:id="@+id/emailInputLayout"
88                      android:layout_width="match_parent"
89                      android:layout_height="wrap_content"
90                      android:hint="E-mail">
```

```
91
92                  <com.google.android.material.textfield.TextInputEditText
93                      android:id="@+id/editText_email"
94                      android:layout_width="match_parent"
95                      android:layout_height="wrap_content"
96                      android:inputType="textEmailAddress" />
97
98              </com.google.android.material.textfield.TextInputLayout>
99
100             <RadioGroup
101                 android:id="@+id/radioGroup_gender"
102                 android:layout_width="match_parent"
103                 android:layout_height="wrap_content"
104                 android:layout_marginTop="24dp"
105                 android:orientation="horizontal">
106
107                 <RadioButton
108                     android:id="@+id/radioButton_male"
109                     android:layout_width="wrap_content"
110                     android:layout_height="wrap_content"
111                     android:text="Male" />
112
113                 <RadioButton
114                     android:id="@+id/radioButton_female"
115                     android:layout_width="wrap_content"
116                     android:layout_height="wrap_content"
117                     android:text="Female" />
118
119             </RadioGroup>
120
121             <com.google.android.material.textfield.TextInputLayout
122                 android:id="@+id/passwordInputLayout"
123                 android:layout_width="match_parent"
124                 android:layout_height="wrap_content"
125                 android:hint="Password">
126
127                 <com.google.android.material.textfield.TextInputEditText
128                     android:id="@+id/editText_password"
129                     android:layout_width="match_parent"
130                     android:layout_height="wrap_content"
131                     android:inputType="textPassword" />
```

```
132
133                 </com.google.android.material.textfield.TextInputLayout>
134
135                 <com.google.android.material.textfield.TextInputLayout
136                     android:id="@+id/confirmPasswordInputLayout"
137                     android:layout_width="match_parent"
138                     android:layout_height="wrap_content"
139                     android:hint="Confirm␣Password">
140
141                     <com.google.android.material.textfield.TextInputEditText
142                         android:id="@+id/editText_confirm_password"
143                         android:layout_width="match_parent"
144                         android:layout_height="wrap_content"
145                         android:inputType="textPassword" />
146
147                 </com.google.android.material.textfield.TextInputLayout>
148
149                 <com.google.android.material.textfield.TextInputLayout
150                     android:id="@+id/contactNoInputLayout"
151                     android:layout_width="match_parent"
152                     android:layout_height="wrap_content"
153                     android:hint="Contact␣No.">
154
155                     <com.google.android.material.textfield.TextInputEditText
156                         android:id="@+id/editText_contact_no"
157                         android:layout_width="match_parent"
158                         android:layout_height="wrap_content"
159                         android:inputType="phone" />
160
161                 </com.google.android.material.textfield.TextInputLayout>
162
163                 <com.google.android.material.button.MaterialButton
164                     android:id="@+id/button_register_submit"
165                     android:layout_width="match_parent"
166                     android:layout_height="wrap_content"
167                     android:layout_marginTop="24dp"
168                     android:backgroundTint="#4D959B"
169                     android:onClick="onClickRegister"
170                     android:text="Register"
171                     android:textColor="#FFFFFF" />
172             </LinearLayout>
```

```
173         </ScrollView>
174     </androidx.cardview.widget.CardView>
175 </androidx.constraintlayout.widget.ConstraintLayout>
```

3. DoctorRegistration.Java

```
1     package com.example.finalehomeopathy;
2
3 import android.content.Intent;
4 import android.os.Bundle;
5 import android.text.InputType;
6 import android.text.TextUtils;
7 import android.util.Log;
8 import android.view.View;
9 import android.widget.EditText;
10 import android.widget.RadioButton;
11 import android.widget.Toast;
12
13 import androidx.appcompat.app.AppCompatActivity;
14
15 import com.google.firebase.auth.FirebaseAuth;
16 import com.google.firebase.auth.FirebaseUser;
17 import com.google.firebase.database.DatabaseReference;
18 import com.google.firebase.database.FirebaseDatabase;
19 import com.google.firebase.messaging.FirebaseMessaging;
20
21 import android.text.InputType;
22
23
24 public class DoctorRegistrationActivity extends AppCompatActivity {
25
26     private EditText editTextName, editTextDesignation, editTextEmail, editTex
27     private RadioButton radioButtonMale, radioButtonFemale;
28     private FirebaseAuth mAuth;
29     private DatabaseReference mDatabase;
30
31     @Override
32     protected void onCreate(Bundle savedInstanceState) {
33         super.onCreate(savedInstanceState);
34         setContentView(R.layout.activity_doctor_registration);
35
36         mAuth = FirebaseAuth.getInstance();
```

```
37          mDatabase = FirebaseDatabase.getInstance().getReference();
38
39          editTextName = findViewById(R.id.editText_name);
40          editTextDesignation = findViewById(R.id.editText_designation);
41          editTextDesignation.setInputType(InputType.TYPE_CLASS_TEXT);
42          editTextEmail = findViewById(R.id.editText_email);
43          editTextPassword = findViewById(R.id.editText_password);
44          editTextConfirmPassword = findViewById(R.id.editText_confirm_password)
45          editTextContactNo = findViewById(R.id.editText_contact_no);
46          radioButtonMale = findViewById(R.id.radioButton_male);
47          radioButtonFemale = findViewById(R.id.radioButton_female);
48
49          findViewById(R.id.button_register_submit).setOnClickListener(new View.
50              @Override
51              public void onClick(View v) {
52                  registerUser();
53              }
54          });
55      }
56
57      private void registerUser() {
58          String name = editTextName.getText().toString().trim();
59          String designation = editTextDesignation.getText().toString().trim();
60          String email = editTextEmail.getText().toString().trim();
61          String password = editTextPassword.getText().toString().trim();
62          String confirmPassword = editTextConfirmPassword.getText().toString()
63          String contactNo = editTextContactNo.getText().toString().trim();
64          String gender = radioButtonMale.isChecked() ? "Male" : "Female";
65
66          if (TextUtils.isEmpty(name) || TextUtils.isEmpty(designation) || TextU
67              Toast.makeText(this, "All␣fields␣are␣required", Toast.LENGTH_SHORT
68              return;
69          }
70
71          if (!password.equals(confirmPassword)) {
72              Toast.makeText(this, "Passwords␣do␣not␣match", Toast.LENGTH_SHORT)
73              return;
74          }
75
76          mAuth.createUserWithEmailAndPassword(email, password)
77                  .addOnCompleteListener(this, task -> {
```

```
                    if (task.isSuccessful()) {
                        FirebaseUser user = mAuth.getCurrentUser();
                        if (user != null) {
                            String userId = user.getUid();
                            DatabaseReference currentUserDb = mDatabase.child
                            currentUserDb.child("name").setValue(name);
                            currentUserDb.child("designation").setValue(desig
                            currentUserDb.child("email").setValue(email);
                            currentUserDb.child("gender").setValue(gender);
                            currentUserDb.child("contactNo").setValue(contact

                            // Save the FCM token if available
                            String fcmToken = FirebaseMessaging.getInstance()
                            if (fcmToken != null) {
                                currentUserDb.child("fcmToken").setValue(fcmTo
                            }

                            Toast.makeText(DoctorRegistrationActivity.this, "
                            startActivity(new Intent(DoctorRegistrationActivit
                            finish();
                        }
                    } else {
                        Toast.makeText(DoctorRegistrationActivity.this, "Regis
                    }
                })
                .addOnFailureListener(e -> {
                    // Log the exception or show a message to the user
                    Log.e("Registration Error", e.getMessage());
                    Toast.makeText(DoctorRegistrationActivity.this, "Registrat
                });
    }
}
```

4. DoctorRegistration.XML

```
    <?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout xmlns:android="http://schem
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:background="#E1F4F5"
```

```
    tools:context=".PatientRegistrationActivity">

    <ImageView
        android:id="@+id/image"
        android:layout_width="154dp"
        android:layout_height="124dp"
        android:layout_centerHorizontal="true"
        android:adjustViewBounds="true"
        android:maxWidth="180dp"
        android:src="@drawable/dashboard"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        tools:ignore="MissingConstraints" />

    <TextView
        android:id="@+id/text1"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="24dp"
        android:text="Welcome␣to␣PQRS␣Homeopathy"
        android:textAlignment="center"
        android:textAppearance="@style/TextAppearance.AppCompat.Headline"
        android:textSize="20dp"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/image" />

    <androidx.cardview.widget.CardView
        android:layout_width="0dp"
        android:layout_height="0dp"
        android:layout_marginStart="16dp"
        android:layout_marginEnd="16dp"
        android:layout_marginTop="16dp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/text1">

        <ScrollView
            android:layout_width="match_parent"
```

```
49              android:layout_height="match_parent"
50              android:fillViewport="true">
51
52              <LinearLayout
53                  android:layout_width="match_parent"
54                  android:layout_height="wrap_content"
55                  android:orientation="vertical"
56                  android:padding="16dp">
57
58                  <com.google.android.material.textfield.TextInputLayout
59                      android:id="@+id/nameInputLayout"
60                      android:layout_width="match_parent"
61                      android:layout_height="wrap_content"
62                      android:hint="Name">
63
64                      <com.google.android.material.textfield.TextInputEditText
65                          android:id="@+id/editText_name"
66                          android:layout_width="match_parent"
67                          android:layout_height="wrap_content"
68                          android:inputType="textPersonName" />
69
70                  </com.google.android.material.textfield.TextInputLayout>
71
72                  <com.google.android.material.textfield.TextInputLayout
73                      android:id="@+id/ageInputLayout"
74                      android:layout_width="match_parent"
75                      android:layout_height="wrap_content"
76                      android:hint="Designation">
77
78                      <com.google.android.material.textfield.TextInputEditText
79                          android:id="@+id/editText_designation"
80                          android:layout_width="match_parent"
81                          android:layout_height="wrap_content"
82                          android:inputType="number" />
83
84                  </com.google.android.material.textfield.TextInputLayout>
85
86                  <com.google.android.material.textfield.TextInputLayout
87                      android:id="@+id/emailInputLayout"
88                      android:layout_width="match_parent"
89                      android:layout_height="wrap_content"
```

```
90                  android:hint="E-mail">
91
92                  <com.google.android.material.textfield.TextInputEditText
93                      android:id="@+id/editText_email"
94                      android:layout_width="match_parent"
95                      android:layout_height="wrap_content"
96                      android:inputType="textEmailAddress" />
97
98              </com.google.android.material.textfield.TextInputLayout>
99
100             <RadioGroup
101                 android:id="@+id/radioGroup_gender"
102                 android:layout_width="match_parent"
103                 android:layout_height="wrap_content"
104                 android:layout_marginTop="24dp"
105                 android:orientation="horizontal">
106
107                 <RadioButton
108                     android:id="@+id/radioButton_male"
109                     android:layout_width="wrap_content"
110                     android:layout_height="wrap_content"
111                     android:text="Male" />
112
113                 <RadioButton
114                     android:id="@+id/radioButton_female"
115                     android:layout_width="wrap_content"
116                     android:layout_height="wrap_content"
117                     android:text="Female" />
118
119             </RadioGroup>
120
121             <com.google.android.material.textfield.TextInputLayout
122                 android:id="@+id/passwordInputLayout"
123                 android:layout_width="match_parent"
124                 android:layout_height="wrap_content"
125                 android:hint="Password">
126
127                 <com.google.android.material.textfield.TextInputEditText
128                     android:id="@+id/editText_password"
129                     android:layout_width="match_parent"
130                     android:layout_height="wrap_content"
```

```
131                     android:inputType="textPassword" />
132
133                 </com.google.android.material.textfield.TextInputLayout>
134
135                 <com.google.android.material.textfield.TextInputLayout
136                     android:id="@+id/confirmPasswordInputLayout"
137                     android:layout_width="match_parent"
138                     android:layout_height="wrap_content"
139                     android:hint="Confirm␣Password">
140
141                     <com.google.android.material.textfield.TextInputEditText
142                         android:id="@+id/editText_confirm_password"
143                         android:layout_width="match_parent"
144                         android:layout_height="wrap_content"
145                         android:inputType="textPassword" />
146
147                 </com.google.android.material.textfield.TextInputLayout>
148
149                 <com.google.android.material.textfield.TextInputLayout
150                     android:id="@+id/contactNoInputLayout"
151                     android:layout_width="match_parent"
152                     android:layout_height="wrap_content"
153                     android:hint="Contact␣No.">
154
155                     <com.google.android.material.textfield.TextInputEditText
156                         android:id="@+id/editText_contact_no"
157                         android:layout_width="match_parent"
158                         android:layout_height="wrap_content"
159                         android:inputType="phone" />
160
161                 </com.google.android.material.textfield.TextInputLayout>
162
163                 <com.google.android.material.button.MaterialButton
164                     android:id="@+id/button_register_submit"
165                     android:layout_width="match_parent"
166                     android:layout_height="wrap_content"
167                     android:layout_marginTop="24dp"
168                     android:backgroundTint="#32cd32␣"
169                     android:onClick="onClickRegister"
170                     android:text="Register"
171                     android:textColor="#FFFFFF" />
```

```
            </LinearLayout>
        </ScrollView>
    </androidx.cardview.widget.CardView>
</androidx.constraintlayout.widget.ConstraintLayout>
```

6.3 User Interface

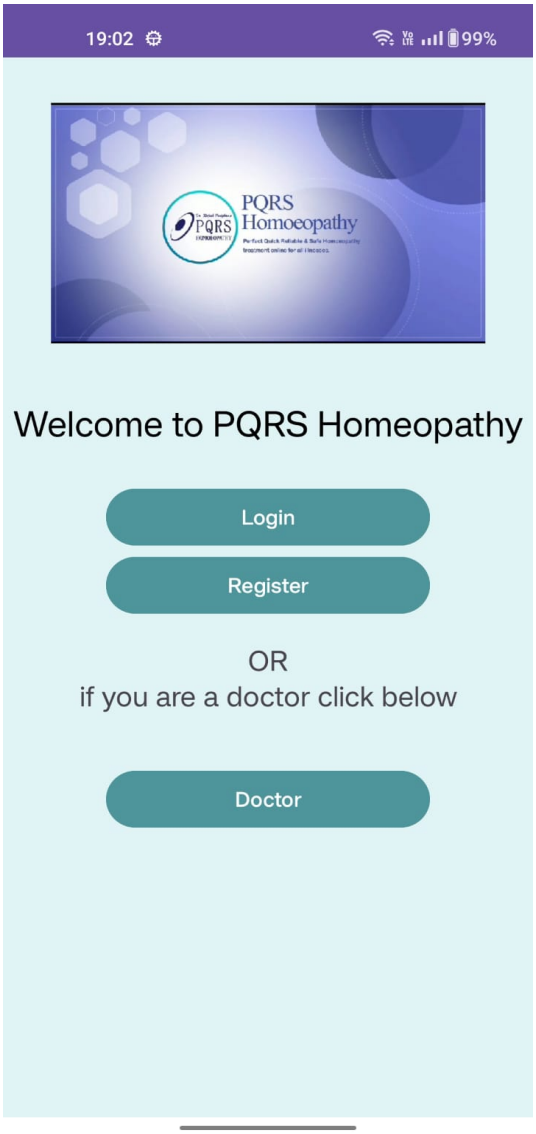


Figure 6.1: Splash Screen

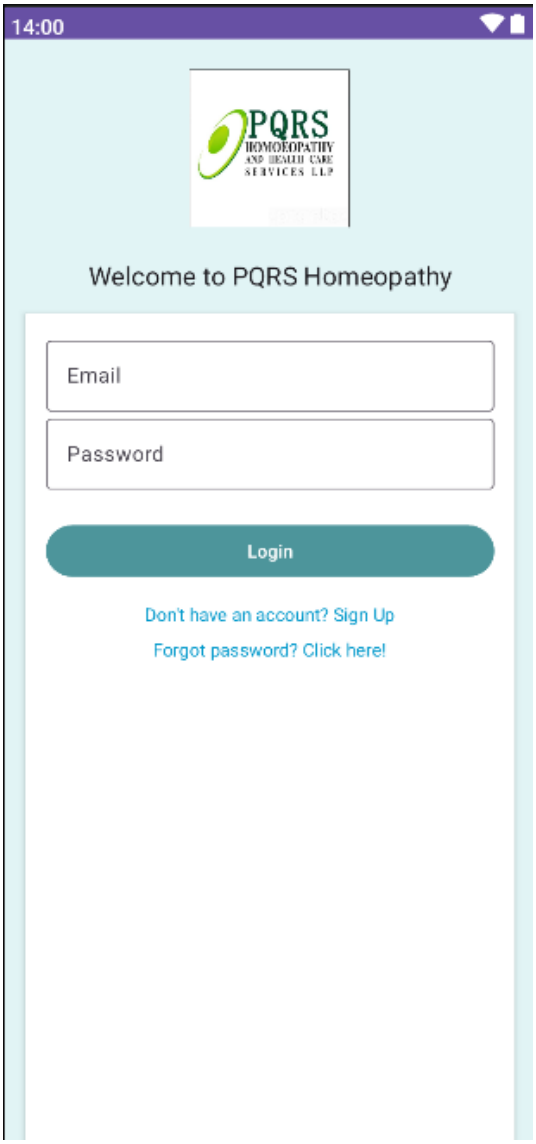


Figure 6.2: Patient Login

Fig.6.1 is the Splash screen which appears immediately when we start our app.It has three buttons, for patients to login and register and for the doctor his login and registration. Fig.6.2 shows the login screen for patients to login wherein the patient once registered enters his credentials and can then use it.

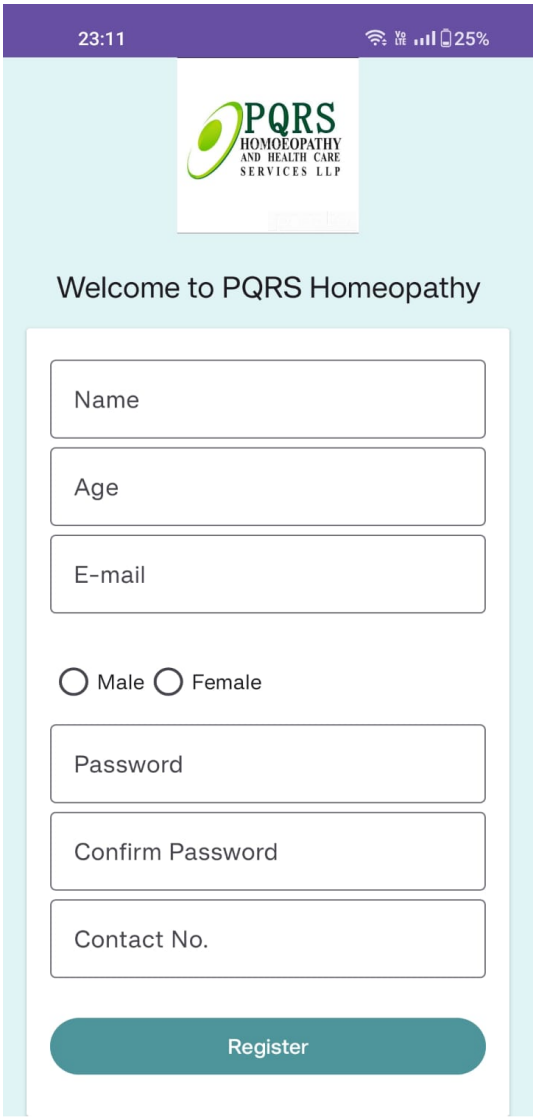


Figure 6.3: Patient Registration

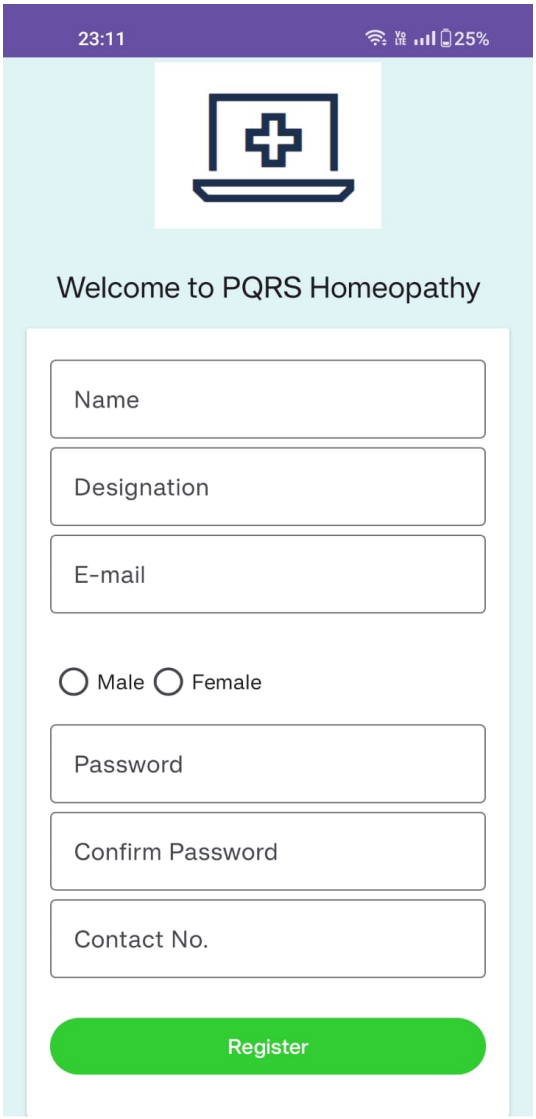


Figure 6.4: Doctor Registration

Fig.6.3 is the patient registration screen wherein the patient registers into the app by entering his/her details. Fig.6.4 is the doctors' registration screen for the doctor wherein he can register in the app to use it.

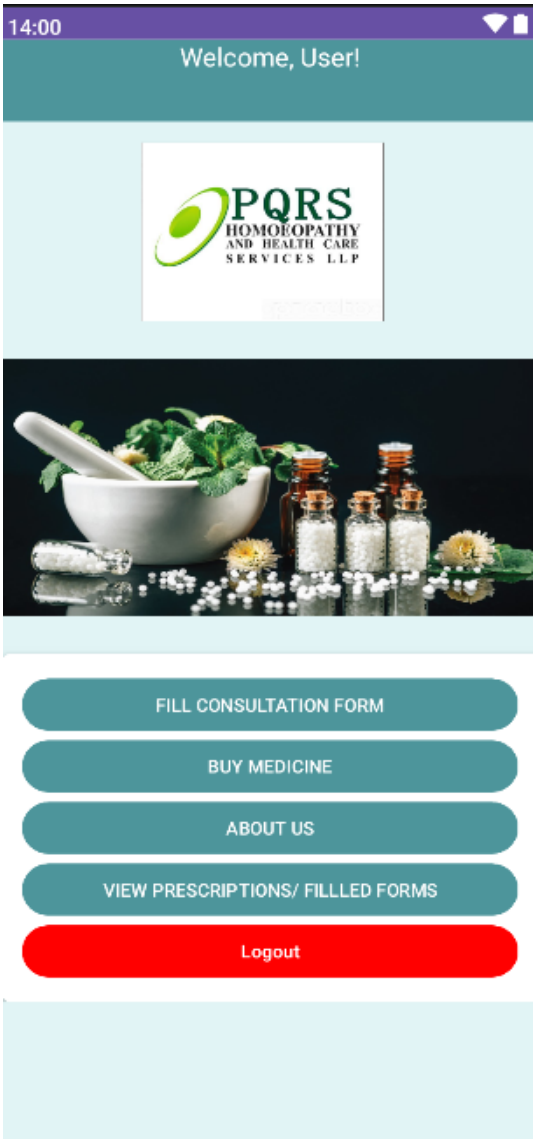


Figure 6.5: Patient Dashboard

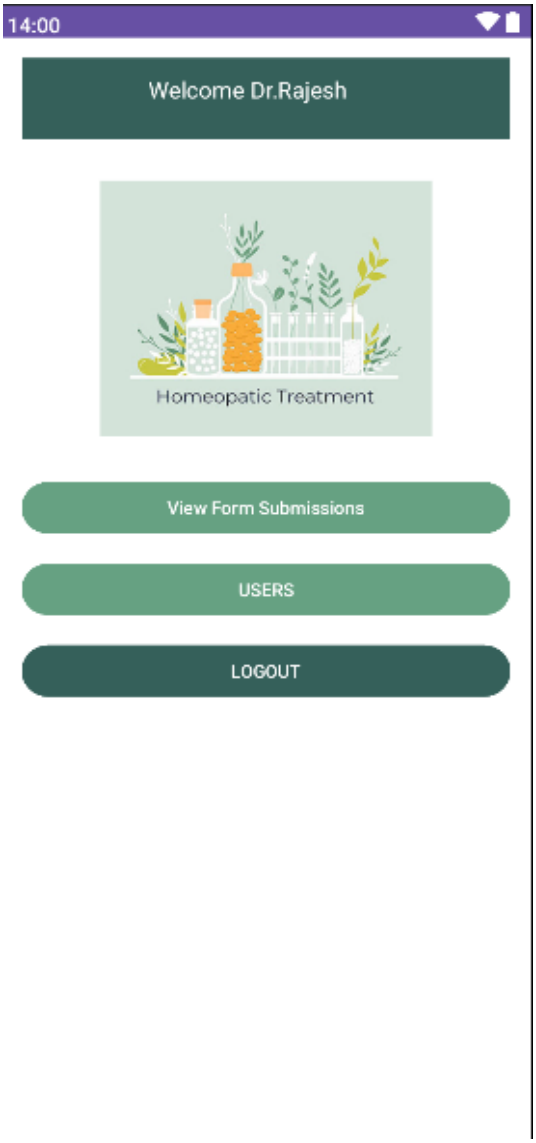


Figure 6.6: Doctor Dashboard

Fig.6.5 is the patient dashboard screen wherein the patient can fill form, view prescriptions, and buy medicine. Fig.6.6 is the doctors' dashboard screen for the doctor wherein he can view his patients, their form responses, and orders.

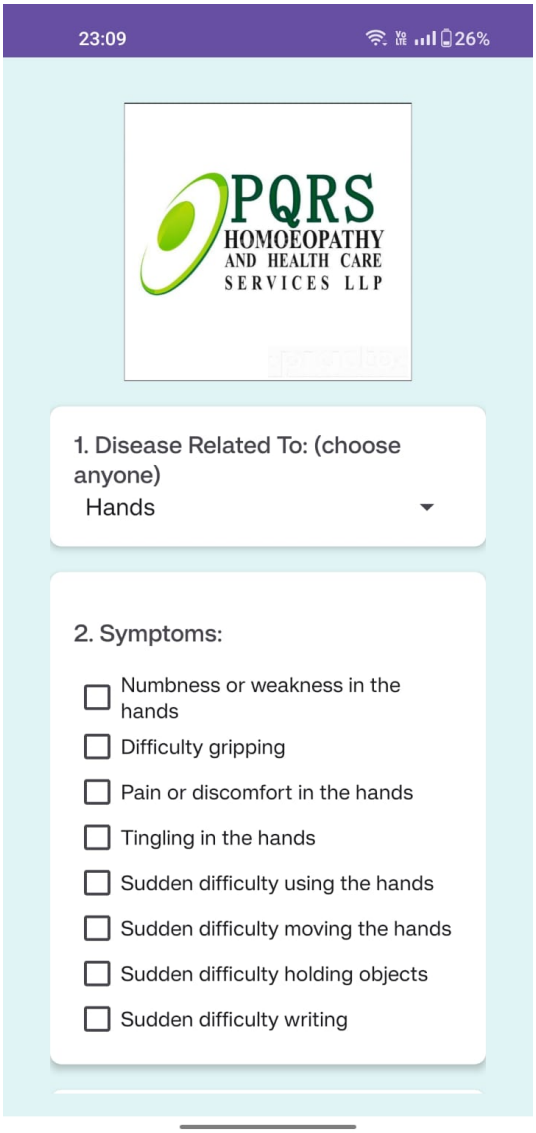


Figure 6.7: Consultation Form

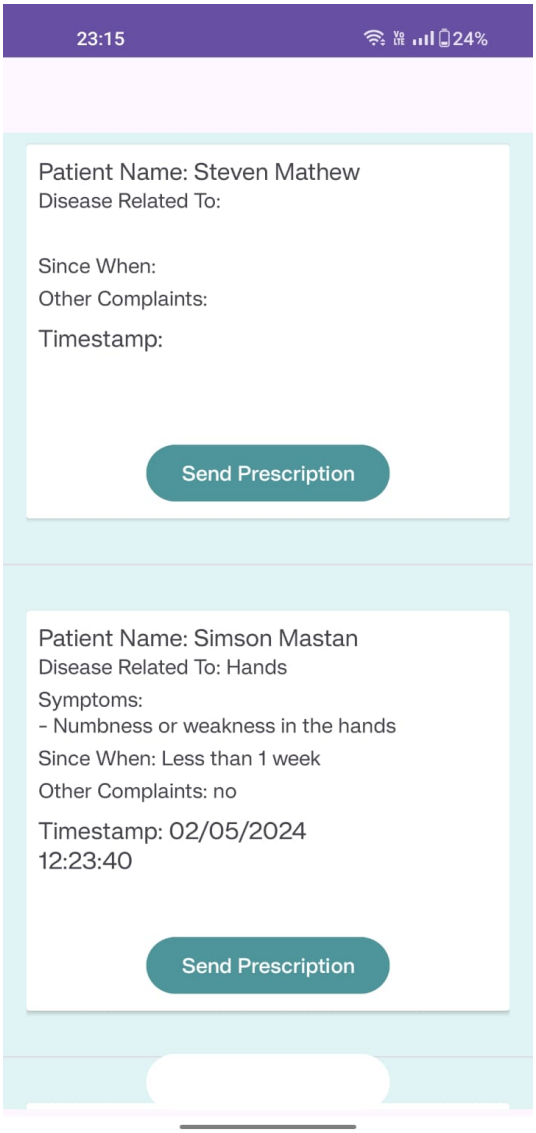


Figure 6.8: Form Responses

Fig.6.7 is the screen showing the consultation form for the patient to fill. Fig.6.8 is the doctor to view form responses sent by the patient based on what he/she is infected from.

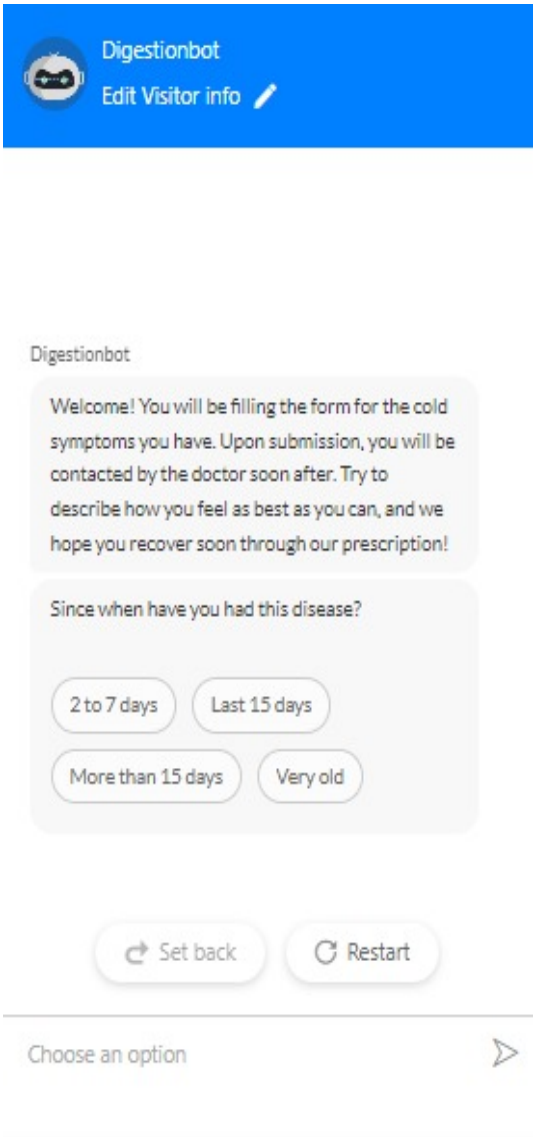


Figure 6.9: Bot Conversation Screen

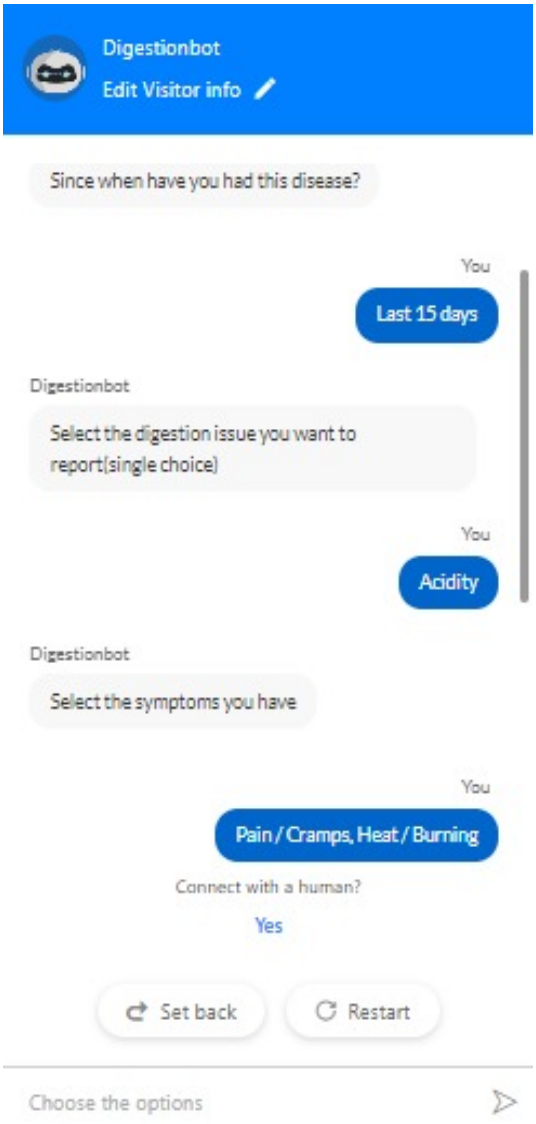


Figure 6.10: Bot Conversation Screen-2

Fig.6.9 and Fig 6.10 are the screens showing how the patient has an interaction with the bot.

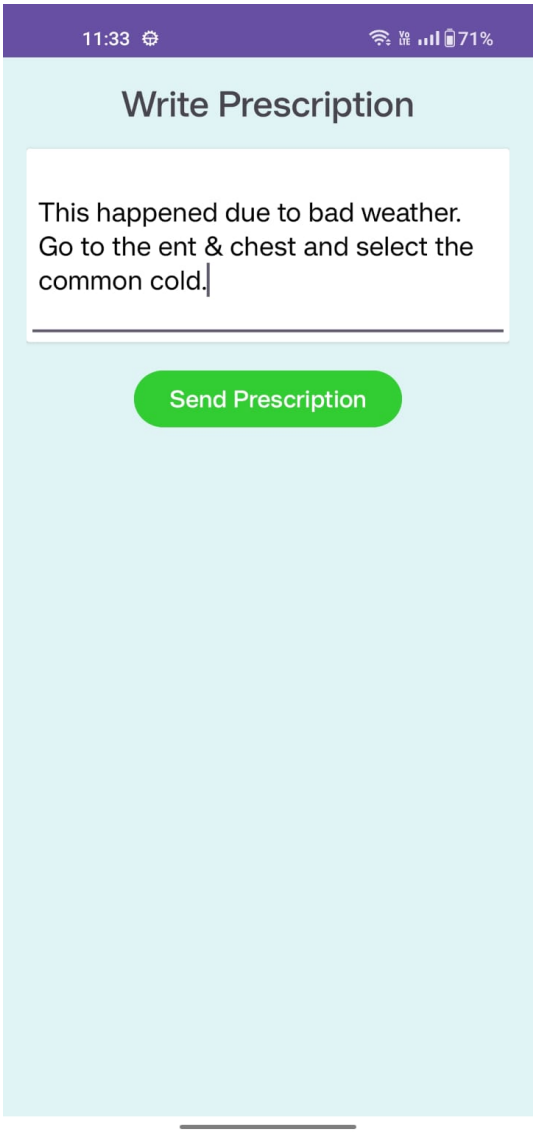


Figure 6.11: Prescription Screen

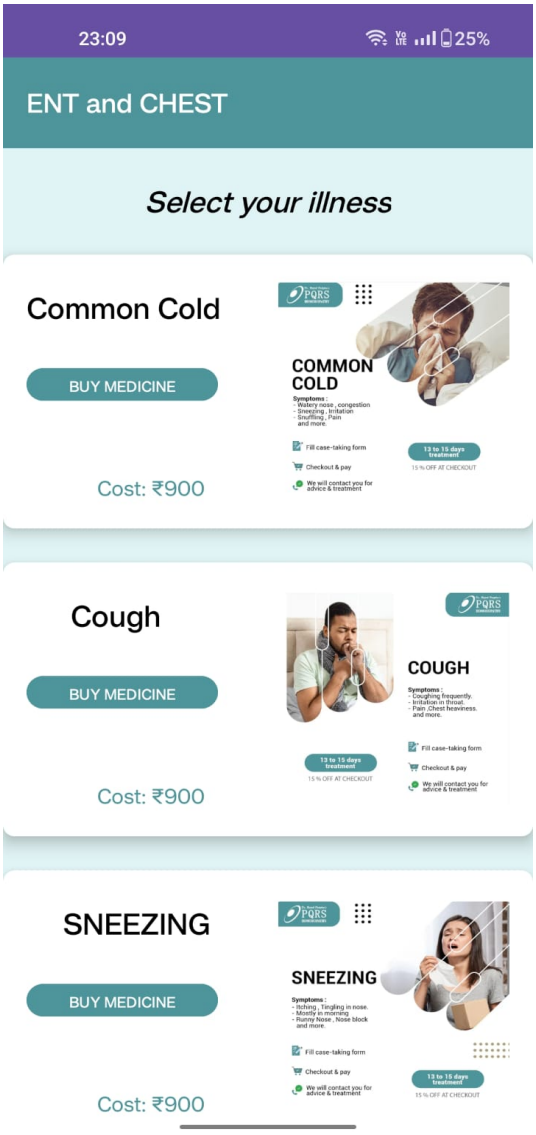


Figure 6.12: E-Commerce Section

Fig.6.11 is the screen showing how the doctor writes a prescription for the patient. Fig.6.12 is the store section wherein the patient buys the medicine prescribed by the doctor.

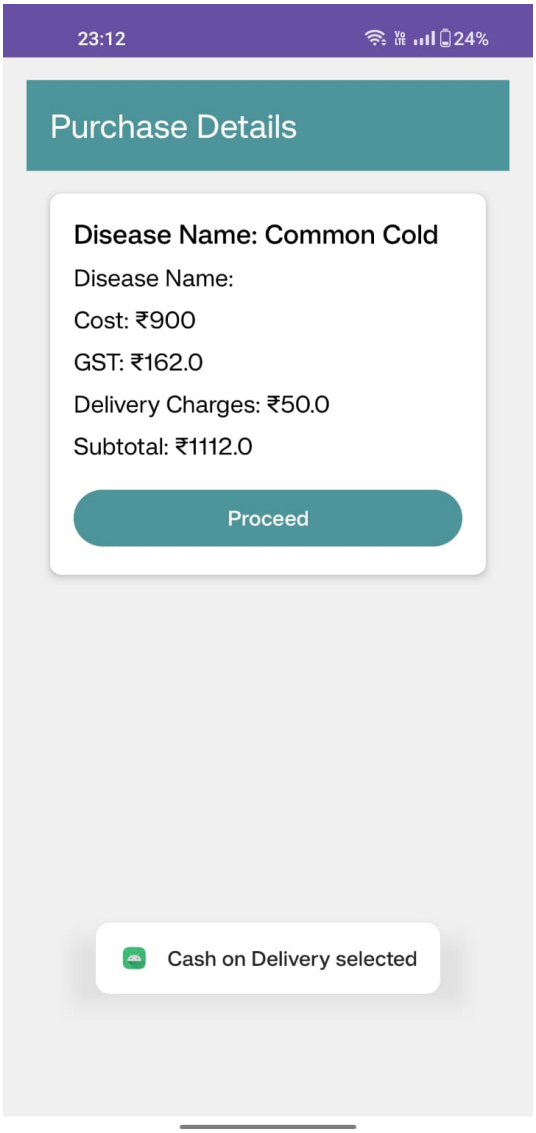


Figure 6.13: Costing Screen

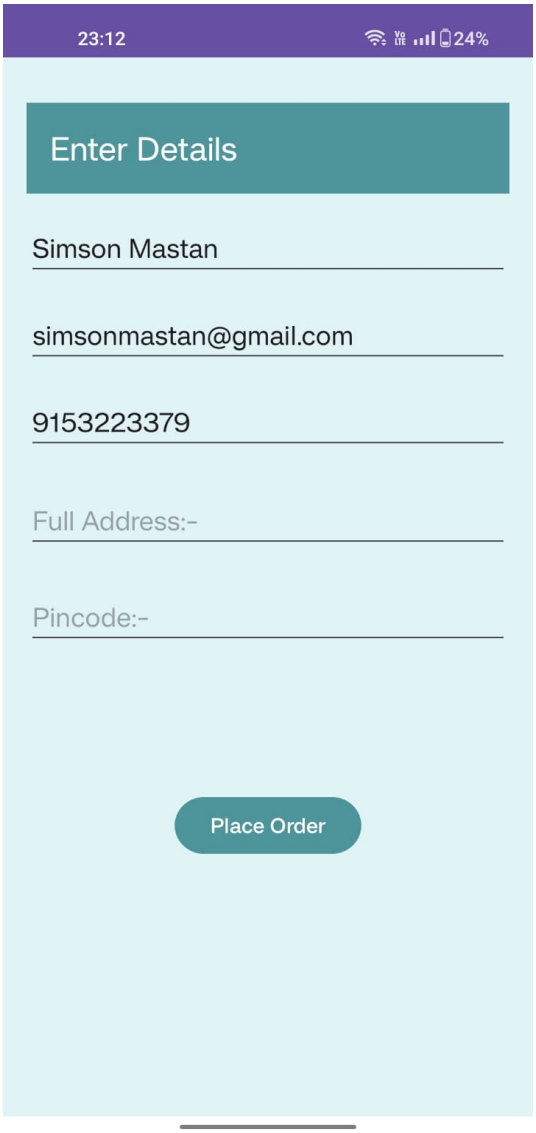


Figure 6.14: Details Section

Fig.6.13 is the screen showing the total cost the patient has to pay for the medicine selected. Fig.6.14 is where the patient fills his/her complete address and then proceeds for the payment.

6.4 Results and discussion

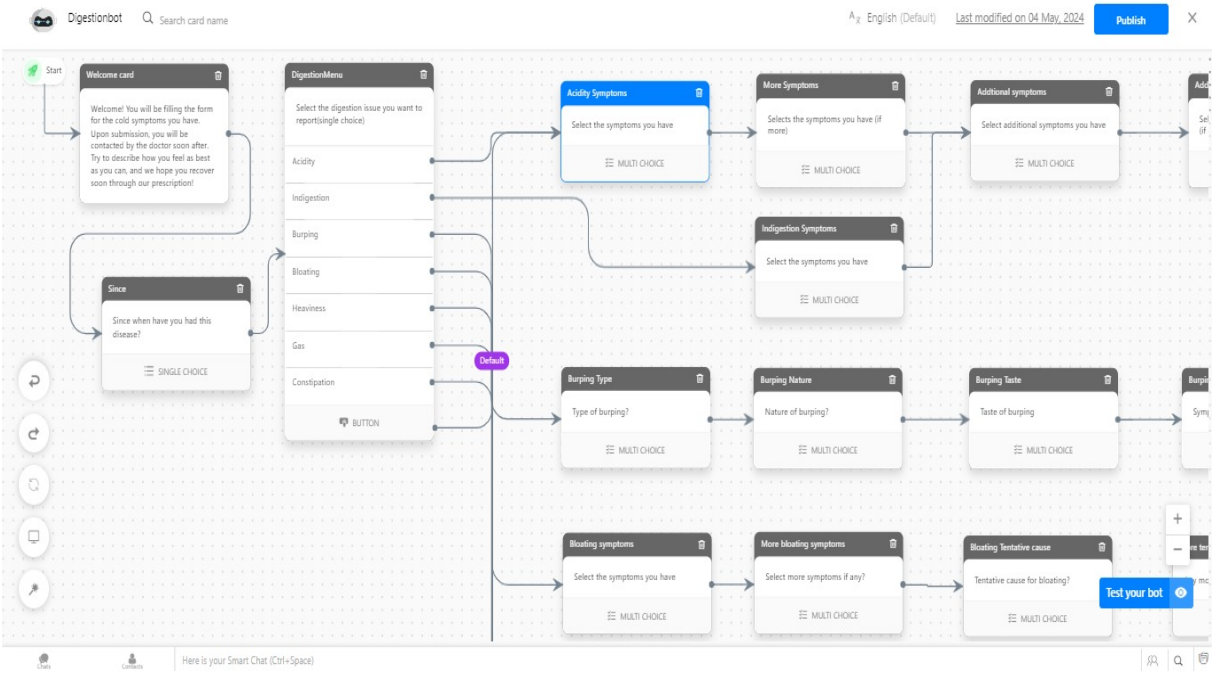


Figure 6.15: Chat-Bot Questionnaire

This figure shows how the data in the chat-bot is hard-coded or is entered manually for the interaction between the patient and the chat-bot. Some of the major fields assigned manually are as follows:

1. ENTDisease
2. ENTSymptoms
3. ENTAdditionalSymptoms
4. ENTAdditionalTypes
5. ENTBetterAfter
6. ENTBetterAfter2
7. ENTDischarges
8. Disease
9. ENTOtherComplaints
10. ENTRespiratoryComplaints
11. ENTRespiratoryComplaints3
12. ENTSeason
13. ENTStillMoreSymptoms

14. ENTSymptoms
15. ENTTentativeCause
16. ENTTiming
17. ENTWorseAfter

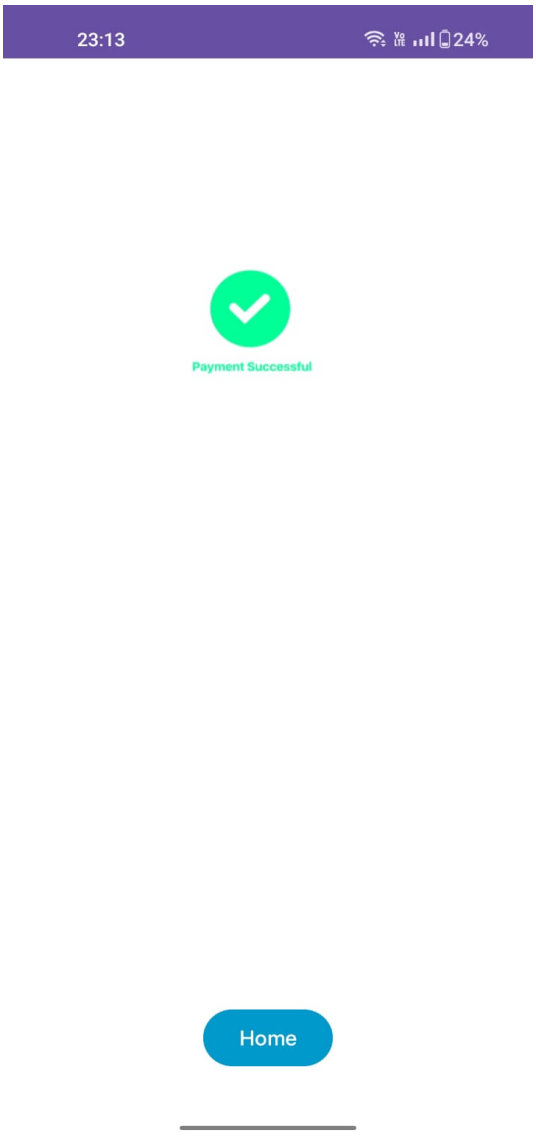


Figure 6.16: Order Placed Screen

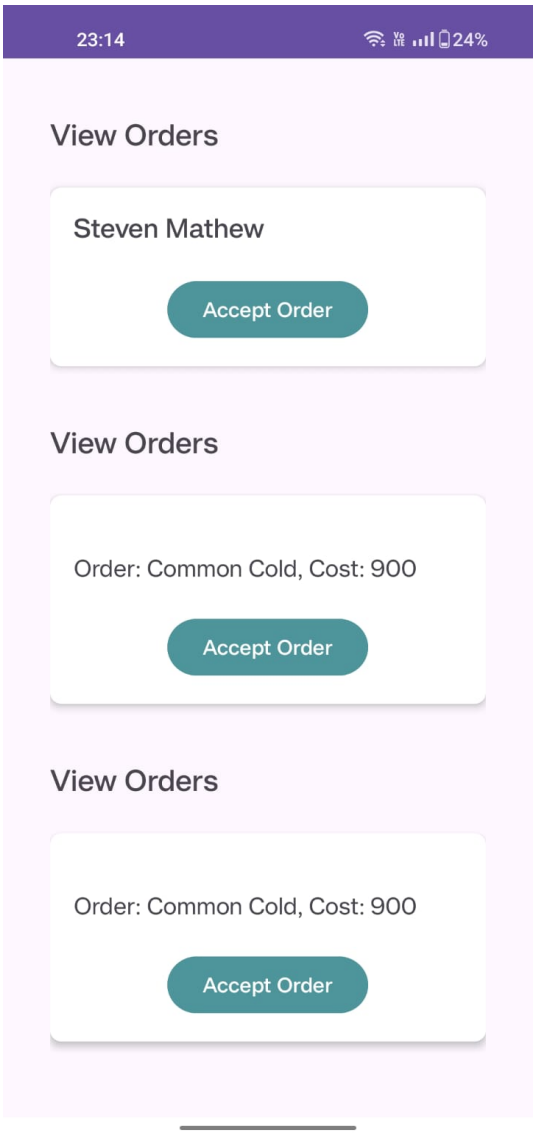


Figure 6.17: Doctor Confirmation

Fig.6.16 shows the final screen from the patients' side is to how the order is placed successfully. Fig.6.17 shows the screen wherein the doctor views the orders placed in the app.

Chapter 7

Conclusion

7.1 Conclusion

In this particular context, we have undertaken an extensive and in-depth analysis of the intricate and nuanced field of homeopathic WissHom standards. Our dedicated efforts have culminated in the meticulous and thoughtful design of a system that is poised for imminent implementation. It brings us great satisfaction to convey that we have not merely conducted a cursory examination but have devoted substantial time and resources to dissecting the issue at hand with laser-like precision. Furthermore, our endeavors have led us to devise a robust and exceptionally ingenious solution, one that stands poised to revolutionize and streamline the often intricate and convoluted processes involved in patient case-processing and consultation. As a result, we anticipate a notable transformation, rendering these processes more accessible, streamlined, and strikingly efficient in their execution.

7.2 Future Scope

i. Enhanced User Interface: Continuously improve the app's user interface to make it more intuitive and user-friendly, enhancing the overall user experience.

ii. Integration of AI and ML: Explore the integration of artificial intelligence (AI) and machine learning (ML) algorithms to provide personalized treatment suggestions based on user input and historical data.

iii. Telemedicine Features: Introduce telemedicine features such as live video consultations between patients and doctors for a more interactive and real-time consultation experience.

iv. Health Monitoring Devices Integration: Integrate with wearable health monitoring devices to collect real-time health data, allowing doctors to make more informed treatment decisions.

v. Medication Reminders and Adherence Tracking: Develop features to remind users to take their medication on time and track medication adherence, promoting better health outcomes.

vi. Expand E-commerce Platform: Expand the e-commerce platform to offer a wider range of homeopathic medicines and health products, catering to diverse user needs.

vii. Feedback and Rating System: Implement a feedback and rating system to gather user feedback on the effectiveness of prescribed treatments and the overall quality of service.

viii. Multilingual Support: Provide multilingual support to cater to users from different regions and linguistic backgrounds, enhancing accessibility and usability.

ix. Data Analytics for Health Insights: Utilize data analytics techniques to derive valuable insights from user data, helping to identify trends, improve treatment efficacy, and optimize resource allocation.

x. Collaboration with Healthcare Providers: Collaborate with healthcare providers and institutions to expand the reach of the app.

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